

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		C (1) reactivity of halogens decreases down the table / chlorine is the most reactive/ iodine is the least reactive (1) award (1) for any of following chlorine displaces bromine and iodine iodine does not displace bromine or chlorine chlorine reacts with sodium bromide and sodium iodide iodine does not react with sodium bromide or sodium iodide	2		1	3		3
		(ii)		$\text{Cl}_2 + 2\text{NaI} \rightarrow 2\text{NaCl} + \text{I}_2$ award (1) for reactants award (1) for products award (1) for balancing - can only be awarded if reactants and products are correct ignore state symbols accept ionic equation $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$				3		
	(b)			$n(\text{Fe}) = \frac{7}{56} = 0.125$ (1) $n(\text{Br}) = \frac{30}{80} = 0.375$ (1) ratio 1:3 therefore FeBr_3 (1) working must be shown		3		3	2	
				Question 10 total	2	6	1	9	2	3